Problem 29.6: Phases in the PMNS Matrix

Source: M. D. Schwartz, Quantum Field Theory and the Standard Model, Problem 29.6.

Show that with general Dirac and Majorana mass matrices, there are three phases in the PMNS matrix, while if the mass matrix is purely Dirac, there is only one. How many phases are there if the masses are purely Majorana?

Solution. A general 3×3 unitary PMNS matrix contains three mixing angles and six phases. The three charged-lepton fields can always be rephased, removing three phases. If the neutrinos are Dirac particles, the three neutrino mass eigenstates can also be rephased. However, one common rephasing of all charged leptons and neutrinos leaves the PMNS matrix unchanged, so only $3 + 3 - 1 = 5$ phases can be removed, leaving one physical phase. If any Majorana mass is present, the neutrino mass eigenstates are generically Majorana and cannot be continuously rephased, since the Majorana condition $N_i^c = N_i$ is preserved only by the discrete transformation $N_i \to \pm N_i$. Consequently, only the three charged-lepton rephasings are available, leaving $6 - 3 = 3$ physical phases. The same reasoning applies to purely Majorana masses. Thus the PMNS matrix contains one physical phase for purely Dirac neutrinos and three physical phases for either Dirac-plus-Majorana or purely Majorana neutrinos.